

GAGAN YALAMURI

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WORK EXPERIENCE

Cantor Fitzgerald • Information Security

Jun 2025 - Present

- > Orchestrated trusted-IP lifecycle across 6+ Azure tenants using XSOAR, GitHub, and Terraform, reducing configuration drift to zero and cutting update time from hours to under 10 minutes.
- > Built Splunk-to-XSOAR containment that publishes blocklists to EDLs and Cloudflare, reducing mean time to contain from ~30 minutes to under 1 minute.
- > Implemented a progressive block policy (15 days then 30 days) with recurrence tracking to harden threat containment and standardize decisions.
- > Delivered 15+ Python utilities and 5+ SOAR playbooks, eliminating ~200 hours per year of manual work (about 4 hours per week).
- > Streamlined SSL certificate renewal automation on F5 BIG-IP via AppViewX APIs, avoiding more than 200 hours per year of manual effort.

NYU Tandon Cybersecurity • Graduate Assistant, Cloud Security

Sept 2025 - Present

- > Designed and taught a hands-on AWS GuardDuty lab for the Cloud Security course, with clear runbooks and teardown steps so students could detect and investigate threats end-to-end.
- > Mentored students across AWS topics (IAM, S3, VPC, EKS, Vault), troubleshooting labs in office hours and online forums and helping them turn cloud-security concepts into working demos.

Secure Systems Lab • Open Source Contributor

Aug 2024 - Nov 2024

- > Standardized release signing with CI checks for projects protecting ~\$5M in assets; prevented 2048-bit fallbacks and unsigned artifacts.
- > Fixed Git and build issues (key-size downgrades, signer trust, merge/submodule drift), shortening release cycles and cutting manual triage time.

Garrett - Advancing Motion • Cybersecurity Intern

Feb 2023 - Aug 2023

- > Unified MISP and OpenCTI for threat intelligence and hunting, automating enrichment and lifting investigation throughput ~30% across SOC workflows.
- > Hardened Mercedes ECU cryptography and prototyped post-quantum X.509 migration; scripted operations for 40+ servers to cut manual work and improve reliability.

HCL Technologies Ltd. • Summer Intern

Jun 2022 - Aug 2022

- > Developed ML-based DDoS prototype (SVM, Logistic Regression), achieving 94% detection accuracy at the SDN layer.
- > Utilized classifiers like SVM to accurately classify over 100,000 simulated IPs, segregating fake addresses with high precision.

PROJECTS

Securing Serverless Applications in AWS (Zero-Trust Framework)

- > Architected Zero-Trust serverless on AWS (Lambda, API Gateway, WAF, Security Hub), enforcing least-privilege IAM and Secrets Manager-based credential protection.
- > Delivered a hardened application with WAF threat mitigation, plus a risk assessment and cloud compliance guide for secure, auditable deployments.

Post-Quantum Cryptography Toolkit (PQCT)

- > Built a Python cryptography toolkit implementing KEM and Dilithium signatures, achieving ~40% speedup via NumPy vectorization and optimized NTT routines.
- > Released on PyPI with a hybrid X.509 prototype for post-quantum PKI, enabling staged migration to quantum-safe encryption and signatures.

Splunk SOAR - Adaptive MFA Enforcement

- > Orchestrated Splunk SOAR playbooks to flag identities lacking multi-factor authentication, send three escalating notices, and auto-disable violators pending security review.
- > Drove 2FA enforcement across 1,000+ profiles, reducing IAM administration overhead while strengthening access governance and overall risk posture.

Offensive Security - Exploit Solver Scripts

- > Authored 30+ pwntools solvers (buffer overflow, ROP, shellcode) standardizing payload templates and helpers to cut time-to-exploit from hours to minutes.

RESEARCH PAPER

- > Yalamuri, G., Honnavalli, P., & Eswaran, S. (2022). A Review of the Present Cryptographic Arsenal to Deal with Post-Quantum Threats. *Procedia Computer Science*, 215, pp.834–845. **Cited 50+ times** (<https://doi.org/10.1016/j.procs.2022.12.086>)

CERTIFICATIONS

- > Certified Network Security Professional by The SecOps Group

EDUCATION

Master of Science in Cybersecurity (GPA: 4.0) • New York University, New York

Sep 2024 - May 2026 (Expected)

Bachelor of Technology in Computer Science and Engineering • PES University, Bangalore

Aug 2019 - July 2023

SKILLS

Programming & Scripting: Python, Go, C/C++, Java, Bash, PowerShell, PHP, SQL, JavaScript

Operating Systems & Cloud: Windows, Kali Linux/Ubuntu, macOS, AWS (EC2, S3, GuardDuty, Lambda), GCP, Azure, Kubernetes (GKE), Docker, OT/ICS, IoT

Red/Purple Team & Post-Exploitation: Cobalt Strike, PowerShell Empire, Covenant, BloodHound, Mimikatz, Binary Ninja, Ghidra, Pwntools, GDB, Strace

Recon & Vulnerability Assessment: Nmap, Nessus, OpenVAS, Burp Suite, OWASP ZAP, SQLMap, AppScan, Metasploit, Wireshark, IDS/IPS

Detection Engineering & SIEM: Splunk, Wazuh, AWS GuardDuty/CloudWatch, EDR, custom log pipelines, XSOAR, Xpanse Cloud Suite, Sentinel

DevSecOps & Automation: Terraform, Ansible, GitHub Actions, SAST / DAST pipelines, CloudFox, Azure (IAM, API integrations, App registrations)

Frameworks & Methodologies: MITRE ATT&CK, PTES, OWASP Top 10, NIST SP 800-53, ISO 27001, Threat Modeling, Risk Assessments